

COURSE OUTLINE – subject to changes :

L-1	Mon	12	Jan	Welcome to MCS 472 – Statistical Reasoning
L-2	Wed	14	Jan	Statistical Distributions – Taguchi Quality Control
L-3	Fri	16	Jan	The Monte Carlo Method – averaging simulations
	Mon	19	Jan	Martin Luther King day – no classes
L-4	Wed	21	Jan	Proposals for Project I, Report Writing
L-5	Fri	23	Jan	Data Analysis with pandas
L-6	Mon	26	Jan	z-transforms and linear recursions
L-7	Wed	28	Jan	filters and stability
L-8	Fri	30	Jan	Bode plots, feedback, decibels
L-9	Mon	2	Feb	Sampling, properties of DFT
L-10	Wed	4	Feb	Filter design and FFT
L-11	Fri	6	Feb	Image processing with FFT
L-12	Mon	9	Feb	Presentations of Project I
L-13	Wed	11	Feb	Presentations of Project I
L-14	Fri	13	Feb	Presentations of Project I
L-15	Mon	16	Feb	Linear Programming
L-16	Wed	18	Feb	Regression
L-17	Fri	20	Feb	Least Squares Chebyshev and Fourier
L-18	Mon	23	Feb	Cost-Benefit Analysis
L-19	Wed	25	Feb	Microeconomics – Macroeconomics
L-20	Fri	27	Feb	review for the midterm exam
L-21	Mon	2	Mar	Midterm Exam
L-22	Wed	4	Mar	supervised and unsupervised Machine Learning
L-23	Fri	6	Mar	Machine Learning with scikit-learn
L-24	Mon	9	Mar	an introduction to Deep Learning
L-25	Wed	11	Mar	face recognition; ranking data
L-26	Fri	13	Mar	ordinary differential equations; the scientific method
L-27	Mon	16	Mar	Presentations of Project II
L-28	Wed	18	Mar	Presentations of Project II
L-29	Fri	20	Mar	Presentations of Project II
L-30	Mon	30	Mar	dynamics in mechanics and biology
L-31	Wed	1	Apr	models of pursuit; multiple mass springs and traffic; nonlinear systems
L-32	Fri	3	Apr	frequency domains, signals and plants
L-33	Mon	6	Apr	cascading plants, circuits, stability
L-34	Wed	8	Apr	filters and feedback
L-35	Fri	10	Apr	control, root locus, Nyquist analysis
L-36	Mon	13	Apr	introduction to partial differential equations
L-37	Wed	15	Apr	separation of variables: solving problems
L-38	Fri	17	Apr	traffic flow modeling
L-39	Mon	20	Apr	periodic steady state; the Helmholtz equation
L-40	Wed	22	Apr	modeling vibrations by Bessel functions
L-41	Fri	24	Apr	review for the final exam
L-42	Mon	27	Apr	Presentations of Project III
L-43	Wed	29	Apr	Presentations of Project III
L-44	Fri	1	May	Presentations of Project III

Final exam: Tuesday 5 May, 8am to 10am, 304 Burnham Hall.